

Principles of Mobile Communication-I

$$K = SN$$

$$C = MSN$$

$$C = MK$$

K = Full system w/ BW

S = Each cell w/ BW

N = Cluster size

M = No. of Replicas

C = Total No. of duplex channel (BW).

Problem: 1

Consider four different cellular systems that share the following characteristics. The frequency bands are 825 to 845 MHz for mobile unit transmission and 870 to 890 MHz for base station transmission. A duplex circuit consists of one 30 kHz channel in each direction. The systems are distinguished by the reuse factor, which is 4, 7, 12 and 19, respectively.

- (a) Suppose that in each of the systems, the cluster of cells (4, 7, 12, 19) is duplicated 16 times. Find the number of simultaneous communications that can be supported by each system.
- (b) Find the number of simultaneous communications that can be supported by a single cell in each system.
- (c) What is the area covered, in cells, by each system?
- (d) Suppose the cell size is the same in all four systems and a fixed area of 100 cells is covered by each system. Find the number of simultaneous communications that can be supported by each system.

$$\Rightarrow N^2 = 42.06$$

$$\Rightarrow N = 6.5$$

$$\approx 7$$

$\therefore$ cluster size = 7.

Problem-3

Assume a system of 32 cells with a cell radius of 1.6 km, a total of 32 cells, a total frequency bandwidth that supports 336 traffic channels and reuse factor of $N=7$. If there are 32 total cells:

(a) What geographic area is covered?

(b) How many channels are there per cell and?

(c) What is the total number of concurrent calls that can be handled?

(d) Repeat for a cell radius of 0.8 km and 128 cells.

Solution:

(a) Here, $R = 1.6$ km

$$\therefore \text{area of each cell} = \frac{3\sqrt{3}}{2} R^2$$

$$\begin{aligned} \therefore \text{Total area} &= 32 \times \frac{3\sqrt{3}}{2} R^2 \\ &= 32 \times \frac{3\sqrt{3}}{2} (1.6)^2 \\ &= 212.8344 \text{ km}^2. \end{aligned}$$

In a cellular system SIR is 18 dB. Find the appropriate cluster size, Assume that path loss factor is 4 and number of co-channel in 1st tier is 6.

Solution:

Here, SIR = 18 dB

$$\Rightarrow \text{SIR}_{\text{dB}} = 18 \text{ dB}$$

Now, $\text{SIR}_{\text{dB}} = 10 \log \text{SIR}_w$

$$\Rightarrow 18 = 10 \log \text{SIR}_w$$

$$\Rightarrow \log \text{SIR}_w = \frac{18}{10}$$

$$\Rightarrow \text{SIR}_w = \text{antilog} \left(\frac{18}{10} \right)$$

$$= 10^{(18/10)}$$

$$= 10^{1.8}$$

$$= 63.096 \text{ watt}$$

Now, we know that,

$$\text{SIR} = \frac{(\sqrt{3}N)^m}{K}$$

$$\Rightarrow 63.096 = \frac{(\sqrt{3}N)^m}{K}$$

$$\Rightarrow 63.096 = \frac{(\sqrt{3}N)^4}{6}$$

$$\Rightarrow (\sqrt{3}N)^4 = 63.096 \times 6$$

$$\Rightarrow (\sqrt{3})^4 \times (N)^4 = 63.096 \times 6$$

$$\Rightarrow N^2 = \frac{63.096 \times 6}{3^2}$$

Here,

$$m = 4$$

$$K = 6$$

(12) # of simultaneous communication supported by each cell =

$$\text{for } N=4, \rightarrow \frac{666}{4} = 166$$

$$\text{for } N=7, \rightarrow \frac{666}{7} = 95$$

$$\text{for } N=12, \rightarrow \frac{666}{12} = 55$$

$$\text{for } N=19, \rightarrow \frac{666}{19} = 35$$

(13) area covered in each system =

$$\text{for } N=4, \rightarrow \text{area in cells} = 16 \times 4 = 64$$

$$\text{" } N=7, \text{" } " = 16 \times 7 = 112$$

$$\text{" } N=12, \text{" } " = 16 \times 12 = 192$$

$$\text{" } N=19, \text{" } " = 16 \times 19 = 304$$

(14) # of simultaneous communication when fixed area is 100 cells -

$$\text{for } N=4, \frac{666}{4} \times 100 = 166 \times 100 = 16600$$

$$\text{" } N=7, \frac{666}{7} \times 100 = 95 \times 100 = 9500$$

$$\text{" } N=12, \frac{666}{12} \times 100 = 55 \times 100 = 5500$$

$$\text{" } N=19, \frac{666}{19} \times 100 = 35 \times 100 = 3500$$

(b) Total channels = 336 ,

cluster size = 7

$$\text{So, \# of channel/cell} = \frac{336}{7}$$

$$= 48$$

(c) From previous math we can say,

a single cell can handle 48 concurrent calls.

$$32 \text{ " " " } (48 \times 32)$$

$$= 1536$$

$$(d) \text{ Total area} = 128 \times \frac{3\sqrt{3}}{2} R^2 = 128 \times \frac{3\sqrt{3}}{2} \times (0.8)^2 = 212.8344 \text{ km}^2$$

$$\text{\# of channel/cell} = \frac{336}{7} = 48$$

$$128 \text{ cell can handle} = 128 \times 48$$

$$= 6144 \text{ concurrent calls.}$$

Problem-4

If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular system, which uses two 25 kHz simplex channels to provide full duplex voice and control channels. Compute the number of channels available per cell if system uses 4 cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels and rest voice channels in each cell for the above system?

Solution:

Given, $N = 4$

$$\begin{aligned}\text{Total BW} &= 33 \text{ MHz} \\ &= 33 \times 10^3 \text{ kHz}\end{aligned}$$

$$\text{Channel BW} = 2 \times 25 = 50 \text{ kHz} \quad (\because \text{Duplex})$$

$$\therefore \text{total \# of channel} = \frac{33 \times 10^3 \text{ kHz}}{50 \text{ kHz}} = 660$$

$$\begin{aligned}\text{Control channel BW} &= 1 \text{ MHz} \\ &= 1 \times 10^3 \text{ kHz}\end{aligned}$$

$$\therefore \text{\# of control channel} = \frac{1 \times 10^3 \text{ kHz}}{50 \text{ kHz}} = 20$$

$$\therefore \text{\# of voice channel} = 660 - 20 = 640$$

$$\therefore \text{\# of control channels per cell} = \frac{20}{4} = 5$$

$$\begin{aligned}\text{\# of voice channels per cell} &= \frac{640}{4} \\ &= 160\end{aligned}$$

Principles of Mobile Communication-II

$$A = \lambda h$$

$$A_u = \lambda H \text{ (Erlangs)}$$

$$A = U * A_u \text{ (Erlangs)}$$

$$A_c = U * A_u / c \text{ (Erlangs)}$$

$$U = \frac{A}{A_u}$$

A = Traffic intensity of a system of U users

A_u = Traffic intensity of each user

A_c = " " " " channel (per channel)

A = Avg # of calls ^{arriving} during avg holding period,
(avg holding time is avg ~~per~~ call ~~rate~~)

λ = mean rate of calls attempted per unit time
(call request rate)

h = mean holding time per successful call
(~~of~~ for successful call - avg holding time)

Problem 1:

An urban area has a population of two million residents. Three competing, trunked mobile network systems A, B and C provide cellular service in this area. System A has 394 cells with 19 channels each, system B has 98 cells with 57 channels each and system C has 49 cells with 100 channels each. Find the numbers of users that can be supported at 2% blocking if each user averages two calls per hour at an average call duration of three minutes. Assuming that all three trunked systems are operated at maximum capacity.

Solution:

System A

$$\# \text{ of cell} = 394$$

$$\# \text{ of channel/cell} = 19$$

System B

$$\# \text{ of cell} = 98$$

$$\# \text{ of channel/cell} = 57$$

System C

$$\# \text{ of cell} = 49$$

$$\# \text{ of channel/cell} = 100$$

Given,

$$Gos = 0.02$$

$$\lambda = 2 \text{ calls/hour}$$

$$H = 3 \text{ mins}$$

$$= \frac{3}{60} = 0.05 \text{ hour}$$

$$\therefore A_u = \lambda H = 2 \times 0.05 = 0.1$$

Now,

system A

$$A = U A_u$$

$$\rightarrow U = \frac{A}{A_u}$$

$$= \frac{12}{0.1} \quad [\text{From table}]$$

$$= 120$$

$$\therefore \text{total \# of subscriber} = 394 \times 120$$
$$= 47280$$

System B

$$U = \frac{A}{A_u} = \frac{46}{0.1} = 460$$

$$\therefore \text{total \# of subscriber} = 460 \times 98$$
$$= 45086$$

System C

$$U = \frac{A}{A_u} = \frac{88}{0.1} = 880$$

$$\therefore \text{total \# of subscriber} = 880 \times 49$$
$$= 43120$$

$$\text{So, total \# of users in the whole area} = 47280 + 45086 + 43120$$
$$= 135486 \text{ (Ans.)}$$

Example 2:

Assuming that each user in a system generates a traffic intensity of 0.2 Erlangs, how many users can be supported for 0.1% probability of blocking in an Erlangs B system for a number of trunked channels equal to 60.

Solution:

System is an Erlang B.

$$A_u = 0.2 \text{ Erlangs.}$$

$$\text{Pr [Blocking]} = 0.001$$

$$C = 60 \text{ channels.}$$

From Erlang B figure,

$$A \approx 40 \text{ Erlangs.}$$

$$\therefore U = \frac{A}{A_u} = \frac{40}{0.2} = 200 \text{ users.}$$

Example 1:

How many users can be supported for 0.5% blocking probability for the following number of trunked channel in a BCC system?
(a) 5 (b) 10 (c) 20. Assumed that each user generates 0.1 Erlangs of traffic.

Solution:

(a) $c = 5$

$$G_{OS} = 0.005$$

$$A_u = 0.1$$

$$\therefore A = 1.13$$

$$\therefore \text{Total users, } U = \frac{A}{A_u}$$

$$= \frac{1.13}{0.1}$$

$$= 11.3 \approx 11 \text{ users,}$$

[Table / Graph থেকে c এর G_{OS} এর value - এর corresponding A এর value নেয়া যায়]

(b) $c = 10$

$$G_{OS} = 0.005$$

$$A_u = 0.1$$

$$\therefore A = 3.96$$

$$\therefore U = \frac{A}{A_u} = \frac{3.96}{0.1} = 39 \text{ users.}$$

(c) $c = 20$

$$G_{OS} = 0.005$$

$$A_u = 0.1$$

$$\therefore A = 11.1$$

$$\therefore U = \frac{A}{A_u} = \frac{11.1}{0.1} = 111 \text{ users.}$$

Problem 2:

Consider a 7-cell system covering an area of 3500 km^2 . The traffic in the seven cells is as follows:

Cell number	1	2	3	4	5	6	7
Traffic (Erlangs)	30.8	66.7	48.6	33.2	38.2	37.8	32.6

Each user generates an average of 0.03 Erlangs of traffic per hour, with a mean holding time of 180 sec. The system consists of a total of 395 channels, and is designed for a grade of service of 0.02.

- Determine the number of subscribers in each cell.
 - Determine the number of calls per hour per subscriber.
 - Determine the number of calls per hour in each cell.
 - Determine the number of channels required in each cell.
- [Hints: You will need to extrapolate using Table 10.3]
- Determine the total number of subscribers.
 - Determine the average number of subscribers per channel.
 - Determine the subscriber density per km^2 .
 - Determine the total traffic (total Erlangs).
 - Determine the Erlangs per km^2 .
 - What is the radius of a cell?

Solution:

Here,

$$A_u = 0.03$$

$$H = 120.5$$

$$= \frac{120}{60 \times 60}$$

$$= 0.03 \text{ hour.}$$

$$\# \text{ of channel} = 395$$

$$GOS = 0.02$$

[a]

$$1 \rightarrow U = \frac{A}{A_u} = \frac{30.8}{0.03} = 1026.6$$

$$2 \rightarrow U = \frac{A}{A_u} = \frac{66.7}{0.03} = 2223.3$$

$$3 \rightarrow U = \frac{A}{A_u} = \frac{48.6}{0.03} = 1620$$

$$4 \rightarrow U = \frac{A}{A_u} = \frac{33.2}{0.03} = 1106.68$$

$$5 \rightarrow U = \frac{A}{A_u} = \frac{38.2}{0.03} = 1273.3$$

$$6 \rightarrow U = \frac{A}{A_u} = \frac{37.8}{0.03} = 1260$$

$$7 \rightarrow U = \frac{A}{A_u} = \frac{32.6}{0.03} = 1086.67$$

[b]

$$A_u = \lambda H$$

$$\Rightarrow \lambda = \frac{A_u}{H}$$

$$= \frac{0.03}{0.03}$$

$$= 1$$

$\therefore$ per hour - 1 per subscriber 1 hr call.

c) For .

$$1 \rightarrow U \times \lambda = 1026.6 \times 1 = 1026.6$$

$$2 \rightarrow U \times \lambda = 2223.3 \times 1 = 2223.3$$

$$3 \rightarrow U \times \lambda = 1620 \times 1 = 1620$$

$$4 \rightarrow U \times \lambda = 1106.68 \times 1 = 1106.68$$

$$5 \rightarrow U \times \lambda = 1273.3 \times 1 = 1273.3$$

$$6 \rightarrow U \times \lambda = 1260 \times 1 = 1260$$

$$7 \rightarrow U \times \lambda = 1086.67 \times 1 = 1086.67$$

d) From table,

$$1 \rightarrow 40$$

$$2 \rightarrow 78$$

$$3 \rightarrow 59$$

$$4 \rightarrow 43$$

$$5 \rightarrow 48$$

$$6 \rightarrow 48$$

$$7 \rightarrow 42$$

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e) From @

$$1 \rightarrow 1026.6$$

$$2 \rightarrow 2223.3$$

$$3 \rightarrow 1620.$$

$$4 \rightarrow 1106.68$$

$$5 \rightarrow 1273.3$$

$$6 \rightarrow 1260$$

$$7 \rightarrow 1086.67$$

9596.53

→ total # of subscribers.

Example - 1

For a parabolic reflected antenna with a diameter of 2m operating at 12 GHz. Find the effective area and antenna gain.

Solution:

Given, $d = 2\text{m}$
 $f = 12\text{ GHz}$
 $= 12 \times 10^9\text{ Hz}$
 $c = 3 \times 10^8\text{ m/s}$
 $\therefore \lambda = \frac{c}{f} = \frac{3 \times 10^8\text{ m/s}}{12 \times 10^9\text{ Hz}} = 0.025\text{ m}$

Now, $A = \pi r^2$
 $= \pi \left(\frac{d}{2}\right)^2$
 $= \pi \left(\frac{2}{2}\right)^2$
 $= \pi$

From table (for π)
 $A_e = 0.56A$
 $= 0.56 \times \pi$
 $= 0.56\pi$

$G = \frac{4\pi A_e}{\lambda^2}$
 $= \frac{4\pi}{(0.025)^2} \times 0.56\pi$

$G = 144.16$

$$N = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}}$$

$$P_m = \frac{|E_m|^2}{377} \cdot A_e$$

$$P_r = \frac{4\pi}{\lambda} \cdot \frac{E_{od}^2}{d^2} \cdot h \cdot h_r$$

$$A_e = \frac{G \lambda^2}{4\pi}$$

$$P_m = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L} \quad ; \quad P_m(d) = P_0 + 20 \log\left(\frac{d}{d_0}\right)$$

$$\alpha_f = \frac{2d^2}{\lambda}$$

$$d = 3.57(\sqrt{k h_1} + \sqrt{k h_2})$$

$$(3 \times 10^8 \text{ ms}^{-1})$$

$$c = \text{speed of light}$$

$$\lambda = \text{wavelength}$$

$$f = \text{carrier frequency}$$

$$A_e = \text{Effective area}$$

$$G = \text{Antenna gain}$$

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

$$f) \text{ average \# of subscribers per channel} = \frac{9596.53}{358} = 26.81.$$

$$g) \text{ subscriber density per km}^2 = \frac{9596.53}{3100} = 3.08.$$

$$h) \text{ total traffic (total Erlangs)} = 30.8 + 66.7 + 48.6 + 33.2 + 38.2 + 37.8 + 32.6 = 287.9.$$

$$i) \text{ Erlangs per km}^2 = \frac{287.9}{3100} = 0.09.$$

$$j) \text{ radius of a cell, } A = \frac{3\sqrt{3}}{2} R^2$$

$$\Rightarrow R^2 = \frac{2}{3\sqrt{3}} \times A$$

$$\Rightarrow R^2 = \frac{2}{3\sqrt{3}} \times (3100/7)$$

$$\Rightarrow R = \sqrt{\frac{2 \times 3100}{3\sqrt{3} \times 7}}$$

$$= 13.05 \text{ km.}$$

Example-2

(a) Find the minimum distance between two antennas LOS transmission if one antenna is 100 m high and other is at ground level.

(b) Suppose receiving antenna is 100 m high. To achieve the same distance how high must be the transmitting antenna.

Solution:

(a) $h_1 = 100 \text{ m}$
 $h_2 = 0$ ($\because$ ground)

$$\begin{aligned}\therefore d &= 3.57 (\sqrt{kh_1} + \sqrt{kh_2}) \\ &= 3.57 (\sqrt{kh_1} + 0) \\ &= 3.57 \times \sqrt{kh_1} \\ &= 3.57 \times \sqrt{4/3 \times 100} \\ &= 41.22 \text{ m.}\end{aligned}$$

(b) $h_2 = 100 \text{ m}$
 $d = 41.22 \text{ m}$

$$\begin{aligned}\therefore d &= 3.57 (\sqrt{kh_1} + \sqrt{kh_2}) \\ \Rightarrow 41.22 &= 3.57 (\sqrt{4/3 \times h_1} + \sqrt{4/3 \times 100}) \\ \Rightarrow h_1 &= 4.644 \times 10^{-7} \text{ m.}\end{aligned}$$

$$[N.B \rightarrow h_2 = 10 \text{ m} \text{ then } h_1 = 46.74 \text{ m} \text{ then}]$$

Example-3:

Find the far-field distance for a circular antenna with maximum dimension of 1m and operating at frequency 900 MHz.

Solution:

Here, $D = 1\text{ m}$

$$d_f = \frac{2D^2}{\lambda}$$

$$f_c = 900\text{ MHz} = 900 \times 10^6\text{ Hz}$$

$$\lambda = \frac{c}{f_c} = \frac{3 \times 10^8\text{ m/s}}{900 \times 10^6\text{ Hz}}$$

$$= \frac{1}{3}\text{ m}$$

$$d_f = \frac{2D^2}{\lambda} = \frac{2(1)^2}{\frac{1}{3}}$$

$$= 6\text{ m}$$

$$\frac{1}{\epsilon} = k$$

$$W_{02} = f_9$$

$$1 = -ic = f_{10}$$

$$m_{01} = 10$$

$$1 = 1$$

$$S = \pi r^2$$

$$f = \frac{1}{2\pi} \frac{d\theta}{dt}$$

$$f = \frac{1}{2\pi} \frac{d\theta}{dt}$$

$$= \frac{1}{2\pi} \times \frac{1 \times 1 \times 1}{1 \times 1000 \times 1000 \times 1000}$$

$$W_{01} = 0.1 \times 812.8$$

$$W_{02} = 0.1 \times 812.8$$

Example-4: (Slide 47 math first ver)

A unit gain antenna with maximum dimension of λ_m produced 50 W power at 900 MHz.

(a) Find the received power in dBm at a free space distance of 100 m from the antenna.

(b) Also find the received power at 10 km distance.

Solution:

(a) Here,

$$d = 100 \text{ m}$$

$$\lambda = \frac{c}{f_c}$$

$$= \frac{3 \times 10^8}{900 \times 10^6}$$

$$= \frac{1}{3} \text{ m}$$

$$\left. \begin{array}{l} f_c = 900 \text{ MHz} \\ = 900 \times 10^6 \text{ Hz} \end{array} \right\}$$

From friss eqn,

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

$$= \frac{50 \times 1 \times 1 \times \left(\frac{1}{3}\right)^2}{(4\pi)^2 \times (100)^2 \times 1}$$

$$= 3.518 \times 10^{-6} \text{ W}$$

$$= 3.518 \times 10^{-3} \text{ mW}$$

$$\left. \begin{array}{l} \lambda = \frac{1}{3} \\ P_t = 50 \text{ W} \\ G_t = G_r = 1 \text{ (Unit gain)} \\ d = 100 \text{ m} \\ L = 1 \text{ (Loss factor)} \\ P_r = ? \end{array} \right\}$$

$$h_t = 50 \text{ m}$$

$$h_r = 1.5 \text{ m}$$

$$(a) \quad L = \frac{\lambda}{4} = \frac{\frac{1}{3}}{4} = \frac{1}{12} = 0.0833 \text{ m} = 8.33 \text{ cm}$$

$$A_e = \frac{G_R \lambda^2}{4\pi} = \frac{1.8 \times \left(\frac{1}{3}\right)^2}{4 \times \pi} = 0.016 \text{ m}^2$$

$$(b) \quad E_R = \frac{2E_0 d_0}{\pi d^2} (2\pi h_t h_r) = \frac{2 \times 10^{-3} \times 10^3}{\frac{1}{3} \times (5 \times 10^3)^2} \times 2 \times \pi \times 50 \times 1.5 = 1.13 \times 10^{-4} \text{ V/m}$$

$$P_R = \frac{|E_R|^2 G_R \lambda^2}{480 \pi^2} \sqrt{\frac{(1+b)(1-b)}{1+b}}$$

$$P_R = \frac{(1.13 \times 10^{-4})^2}{377} \times \frac{1.8 \times \left(\frac{1}{3}\right)^2}{4\pi} \sqrt{\frac{1}{2}} = 5.4 \times 10^{-13} \text{ W}$$

$$= \frac{(1.13 \times 10^{-4})^2}{377} \times \frac{1.8 \times \left(\frac{1}{3}\right)^2}{4\pi} \sqrt{\frac{1}{2}} = 5.4 \times 10^{-13} \text{ W}$$

$$= 5.4 \times 10^{-13} \text{ W} = -122.68 \text{ dBW}$$

$$= -122.68 \text{ dBW}$$

$$= -92.68 \text{ dBm}$$

$$10 \log(5.4 \times 10^{-13}) \text{ dBW} = -122.68 \text{ dBW}$$

$$10 \log(1000 \times 5.4 \times 10^{-13}) \text{ dBm} = -92.68 \text{ dBm}$$

Example - 5:

A mobile located 5 km away from a base station and uses vertical $\frac{\lambda}{4}$ monopole antenna with gain 2.55 dB to receive cellular radio signal. The E-field at 1 km from the transmitter is measured to 10^{-3} V/m. The carrier frequency used for this signal is 900 MHz.

(a) Find the length and effective area of the receiving antenna.

(b) Find the received power at distance 5 km using two-ray ground reflection model assuming the height of the transmitting antenna is 50 m and receiving antenna is 1.5 m above the ground.

Solution:

Here, Given

$$d = 5 \text{ km} = 5 \times 10^3 \text{ m}$$

$$E_0 = 10^{-3} \text{ V/m}$$

$$d_0 = 1 \text{ km} = 10^3 \text{ m}$$

$$f_c = 900 \text{ MHz} = 900 \times 10^6 \text{ Hz}$$

$$\begin{aligned} \lambda &= \frac{c}{f_c} = \frac{3 \times 10^8}{900 \times 10^6} \\ &= \frac{1}{3} \text{ m} \end{aligned}$$

$$\begin{aligned} G_R &= 2.55 \text{ dB} \\ &= 1.8 \text{ W} \end{aligned}$$

$$2.55 = 10 \log G_R(\text{W})$$

$$\Rightarrow \frac{2.55}{10} = \log G_R(\text{W})$$

$$\begin{aligned} \Rightarrow G_R(\text{W}) &= \text{Antilog}\left(\frac{2.55}{10}\right) \\ &= 1.8 \text{ W} \end{aligned}$$

$$P_n(d) \text{ dBm} = 10 \log \left[\frac{P_n(d_0)}{0.001 \text{ W}} \right] + 20 \log \left(\frac{d_0}{d} \right)$$

Now,

$$P_n(d) \text{ dBm} = 10 \log P \text{ (mW)}$$

$$= 10 \log (3.5 \times 10^{-3})$$

$$= -24.5 \text{ dBm}$$

Fig. 1.10: Power density at a distance d from a transmitter.

$$d = 10 \text{ km} = 10,000 \text{ m}$$

$$P_0 = -24.5 \text{ dBm} \quad [\text{from previous math}]$$

$$d_0 = 100 \text{ m}$$

$$P_n(d) = P_0 \left(\frac{d_0}{d} \right)^2$$

$$\Rightarrow \log P_n(d) = \log P_0 + 2 \log \left(\frac{d_0}{d} \right)$$

$$\Rightarrow P_n(d) = P_0 + 20 \log \left(\frac{d_0}{d} \right)$$

$$\Rightarrow P_n(d) = -24.5 + 20 \log \left(\frac{100}{10,000} \right)$$

$$= -64.5 \text{ dBm}$$

$$P_n(d) = P_0 + 20 \log \left(\frac{d_0}{d} \right)$$

$$P_n(d) = -24.5 + 20 \log \left(\frac{100}{10,000} \right)$$

$$= -64.5 \text{ dBm}$$

$$P_n(d) = P_0 + 20 \log \left(\frac{d_0}{d} \right)$$

$$P_n(d) = -24.5 + 20 \log \left(\frac{100}{10,000} \right)$$

$$= -64.5 \text{ dBm}$$

$$P_n(d) = P_0 + 20 \log \left(\frac{d_0}{d} \right)$$

$$P_n(d) = -24.5 + 20 \log \left(\frac{100}{10,000} \right)$$

$$= -64.5 \text{ dBm}$$

Example - 6

Compute the diffraction loss for 3 cases :

Assume $\lambda = \frac{1}{3}$ m

$$d_1 = 1 \text{ km}$$

$$d_2 = 1 \text{ km}$$

(a) $h = 25 \text{ m}$

(b) $h = 0 \text{ m}$

(c) $h = -25 \text{ m}$

For each of the three cases, identify the fresnel zones with which the tip of the obstruction lies.

Solution:

Here, $d_1 = d_2 = 1 \text{ km} = 1000 \text{ m}$

(a) $h = 25 \text{ m}$

$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}}$$

$$= 25 \sqrt{\frac{2(1000 + 1000)}{\frac{1}{3} \times 1000 \times 1000}}$$

$$= 2.74$$

for, $v > 2.4$

$$G_d(\text{dB}) = 20 \log \left(\frac{0.255}{v} \right)$$

$$= 21.7 \text{ dB}$$



$$\Delta = \frac{h^2}{2} \left(\frac{d_1 + d_2}{d_1 d_2} \right) = \frac{(25)^2}{2} \times \frac{(1000 + 1000)}{1000 \times 1000}$$

$$= 0.625 \text{ m}$$

$$\Delta = \frac{n\lambda}{2}$$

$$\Rightarrow n = \frac{2\Delta}{\lambda}$$

$$= \frac{2 \times 0.625}{\frac{1}{3}}$$

$$= 3.75 \approx 3$$

so, tip of obstruction completely block the first three fresnel zone.

(b) $h = 0$

$\therefore v = 0$

$$G_d(\text{dB}) = 20 \log(0.5 - 0.62v)$$

$$= 6 \text{ dB}$$

$$[-1 \leq v \leq 0]$$

so, middle of the first fresnel zone.

$$\Delta = 0$$

$$n = 0$$

[Power loss negligible]

(c)

$$h = -25$$

$$v = -2.74$$

((b) is just (-) 25)

$$v < -2.4$$

$$\therefore G_d(\text{dB}) = 0$$

$$\Delta = 0.625 \text{ (a) is so)}$$

$$n = 3.75 \approx 3 \text{ (a) is so)}$$

like (a) but

the obstruction is below the line of sight, (h negative)

Solution:

(a) The frequency band for $MU = 845 - 825 = 20 \text{ MHz}$,
 $BB = 890 - 870 = 20 \text{ MHz}$.

So, $MU \xrightleftharpoons[20\text{ MHz}]{20\text{ MHz}} GS$

So, system BW = $20 + 20 = 40$ MHz (~~400 MHz~~).

Each duplex circuit consists of 30 km²

$\Delta\phi$, Each Channel QW = ~~30 kHz~~ (2x30 kHz)
= 60 kHz ($\therefore$ duplex)

Now, # of channel for the system = $\frac{40 \text{ MHz}}{60 \text{ kHz}}$

$$= \frac{40 \times 10^3 \text{ kHz}}{60 \text{ kHz}}$$

666 channels :-

$M = 16$ (# of duplications)

$$\therefore \# \text{ of channel after replication} = 16 \times 666$$

$$= 20,656.$$

of simultaneous communication = 10,656